Interactions of Maintenance and Production Planning for Multipurpose Process Plants—A System Effectiveness Approach

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Process design and operations are typically concerned with the optimal selection, design, and efficient utilization of assets and resources over a time horizon of interest. The degree of utilization of process plant components critically depends on the availability level of equipment components. These availability levels are mainly associated with equipment reliability, failure characteristics, and maintenance policies. This paper presents a system effectiveness optimization framework for multipurpose plants that involves a novel preventive maintenance model coupled with a multiperiod planning model. This provides the basis for simultaneously identifying production and maintenance policies by exploiting (i) opportunities to perform preventive maintenance during equipment idle time and (ii) flexibility in selecting different production routes to potentially mitigate the adverse effects of equipment failure on the production plan. Extensions to include design decisions are also discussed. All problems are formulated as mixed integer linear programming (MILP) models, and their applicability is demonstrated by a number of illustrative examples.

1. Introduction

The key characteristic of multipurpose process plants is that different products, or even different batches of the same product, can follow different production routes using different units in the plant.[1] To organize the timely production of the required amounts of products at minimum cost, a number of planning and scheduling frameworks can be introduced to handle the allocation of utilities, resources, and production tasks. The degree of utilization of assets and resources, however, is critically associated with the level of availability of equipment components, which is determined by the initial reliability characteristics and the implemented maintenance policy. In such a multipurpose operating environment, the making of maintainability decisions such as the timing of maintenance must account for maintenance opportunities arising from the fact that equipment items can often be left idle because of the production pattern. In addition, flexibility in selecting different production patterns can significantly mitigate the adverse effects of equipment failure on the production process.

In this respect, the problems of determining the optimal maintenance and production policies clearly depend on each another (see Figure 1). If a production plan is fixed and used as an input to an optimization formulation for the determination of an optimal maintenance policy, it is likely that a different production plan would facilitate a better maintenance policy. On the other hand, if a maintenance schedule is fixed and used as input for the determination of an optimal production plan, it is likely that a different maintenance policy would facilitate a better production plan. To overcome these concerns and to quantify the interactions between production and maintenance planning models, proper linking mechanisms between the two models must be established so that a simultaneous strategy can be developed.

The importance of considering reliability and maintenance criteria in process manufacturing, design, and operation has been recognized over the past 15 years (see, for example, refs 2−11). Most of the previous work focused on continuous processes, with only a few works specifically concerned with multipurpose operation (for example, refs 12−16). A common theme that is emerging from previous work is the need for the introduction of consistent and rigorous system effectiveness criteria (for example, refs 10, 17, and 18) to characterize the performance of a process system from both the availability and the productivity points of view.

In this work, we propose a system effectiveness optimization framework for the simultaneous optimization of maintenance and production planning for multipurpose batch plants. The key elements of our approach are (i) an aggregate production planning model that describes the process-related characteristics over a long time horizon of operation; (ii) a maintenance model that describes the reliability characteristics of the system and the effect of maintenance policies on the availability of the equipment components; (iii) linking variables that provide the mechanism for the quantification of the interactions between production and maintenance planning by associating the utilization of process assets and resources with the availability of equipment as determined by the maintenance model; and (iv) an explicit system effectiveness criterion, used as an objective function, that provides a balance between the increased revenues due to extra equipment availability and the increased maintenance costs.

This paper is organized as follows. First, a brief description of the aggregate multiperiod production model adopted here is given. Then, a novel maintenance planning model is proposed for modeling equipment

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deterioration and preventive maintenance activities in a multipurpose process operating environment. The maintenance planning model is then coupled with the multiperiod production planning model into a system effectiveness optimization formulation. Design aspects are also considered. Finally, two numerical examples are presented to demonstrate the applicability of the simultaneous approach.

2. An Aggregate Multiperiod Production Planning Model

In this section, an aggregate multiperiod production planning model is presented based on a state−task network (STN) process representation[19] in which a number of time periods, t, of equal duration, H, are considered.

First, a list of the notation used in the multiperiod planning mathematical model is provided.

Indices

i = processing tasks
j = equipment units
t = time periods
s = states of material
u = utilities

Sets

$\underline{S}_i$ = set of states consumed by task i
$\bar{S}_i$ = set of states produced by task i
$\underline{T}_s$ = set of tasks receiving material from state s
$\bar{T}_s$ = set of tasks producing material in state s
I_j = set of tasks for which unit j is suitable
K_i = set of equipment items suitable for task i

Parameters

ρ_{is} = proportion of input of task i coming from state $s \in \underline{S}_i$
$\bar{\rho}_{is}$ = proportion of output of task i going into state $s \in \bar{S}_i$
p_i = setup and processing time of task i
$\beta_{uij\omega}$ = fixed demand factor for utility u by task i performed in unit j at time ω relative to the start of the task
$\delta_{uij\omega}$ = variable demand factor for utility u by task i performed in unit j at time ω relative to the start of the task
V_j = capacity of unit j
$A_{ut}^{\max}$ = upper bound on the total availability level of utility u during period t
H = duration of each period
$\phi_{ij}^{\min}$ = minimum utilization factor
$\phi_{ij}^{\max}$ = maximum utilization factor

The following key continuous variables are required.

Variables

N_{ijt} = number of batches of task i processed in unit j over time period t
S_{st} = amount of material in state s in storage at the end of period t
D_{st} = amount of material delivered to external customers from state s over period t
B_{ijt} = amount of material undergoing task i in unit j during period t
U_{jt} = expected uptime of unit j during period t

The basic constraints of the aggregate multiperiod planning model are the following:

Resource Utilization Constraints

$$\sum_{i \in I_j} p_i N_{ijt} \le U_{jt} \quad \forall j, t \tag{1}$$

Capacity Constraints

$$\phi_{ij}^{\min} V_j N_{ijt} \le B_{ijt} \le \phi_{ij}^{\max} V_j N_{ijt} \quad \forall i, j \in K_i, t \tag{2}$$

Material Balance Constraints

$$S_{st} = S_{s,t-1} + \sum_{i \in \bar{T}_s} \sum_{j \in K_i} \bar{\rho}_{is} B_{ijt} - \sum_{i \in \underline{T}_s} \sum_{j \in K_i} \rho_{is} B_{ijt} - D_{st} \quad \forall s, t \tag{3}$$

Demand Constraints

$$D_{st}^{\min} \le D_{st} \le D_{st}^{\max} \quad \forall s, t \tag{4}$$

Utility Constraints

$$\sum_i \sum_{j \in K_i} \sum_{\omega=0}^{p_i - 1} \beta_{uij\omega} N_{ijt} + \delta_{uij\omega} B_{ijt} \le A_{ut}^{\max} H \quad \forall u, t \tag{5}$$

The resource utilization constraints in eq 1 ensure that the total processing time on a unit cannot exceed the expected uptime of the unit, while the capacity constraints (eq 2) suggest that batch sizes are considered to vary between the minimum and maximum values. The material balances (eq 3) state that the amount of material in state s at the end of period t is equal to the amount in storage from the previous period, plus the amount produced, and minus the amount consumed and delivered. Finally, the demand constraints (eq 4) state that the demand fluctuates between its lower and upper bounds, while the utility constraints (eq 5) ensure that the utilization levels of utilities such as steam, water, etc., do not exceed the corresponding availability levels.

3. A General Mathematical Model for the Preventive Maintenance of Multipurpose Plants

In this section, an analytical preventive maintenance model is developed, assuming that all multipurpose equipment units are in the wear-out phase, i.e., their failure rate is increasing with time. A set of constraints is proposed to describe the impact of preventive maintenance activities on the equipment failure rate.

The underlying assumptions of the proposed maintenance policy are as follows: (i) The equipment failure rate is (linearly or nonlinearly) increasing with time, but it remains constant within each time period t. (ii) Each preventive maintenance activity restores the component to an as-good-as−new (AGAN) status. (iii) In the event of failure, minimal repair is performed to restore the failed unit to an as-good-as-old (AGAO) status. (iv) According to equipment specifications, preventive maintenance should be performed to each unit j at least every τ_j time periods, i.e., each unit cannot operate for more than τ_j time periods without being maintained.

The notation used in the maintenance mathematical model is listed below:

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θ = number of periods elapsed since unit j was last maintained

Parameters

Δ_j^{c} = corrective maintenance (repair) duration of unit j
Δ_j^{p} = preventive maintenance duration of unit j
τ_j = maximum number of consecutive elapsed time periods without maintenance of unit j

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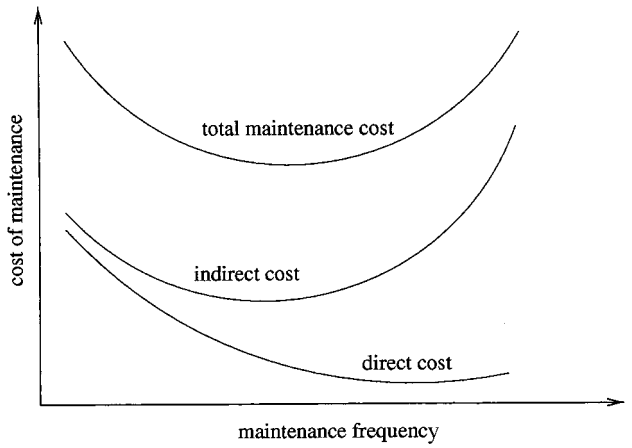


Figure 1. Direct vs indirect maintenance costs.

$\gamma_{j\theta}$ = failure rate value for unit j when the last maintenance action took place θ time periods ago

The following variables are also introduced:

Binary Variables

$X_{jt} = 1$ if preventive maintenance is performed on unit j during period t

0 otherwise

$Z_{jt\theta} = 1$ during period t if unit j was maintained for the last time θ periods ago

0 otherwise

Continuous Variables

λ_{jt} = failure rate of unit j during period t

3.1. Failure Rate Modeling Constraints. Equipment failure rates over time are described as piecewise (nondecreasing) constant functions. In particular, it is assumed that, during each operating period, the unit failure rate has a constant value. This value increases with the number of time periods (wear-out phase) until a preventive maintenance action is performed and restores the state of the unit to its initial condition (AGAN). It is also assumed that the equipment failure rate depends on the time since last repair and not the use made of the equipment.

As stated in the maintenance policy assumptions, the maximum number of periods before a preventive maintenance action is performed, and therefore the maximum number of periods for which the failure rate can be continuously increasing, is equal to τ_j. The constant failure rate value of unit j during each period θ, $1 \le \theta \le \tau_j$, is denoted by $\gamma_{j\theta}$. This suggests that the failure rate of each unit j in the special case when no preventive maintenance activity is performed, other than the compulsory maintenance activities performed every τ_j time periods, is described by

$$\lambda_{jt} = \gamma_{j,1}, \quad t = 1$$

$$\lambda_{jt} = \gamma_{j\theta}, \quad \forall t, \theta = [(t-1) \bmod \tau_j] + 1$$

The variation of failure rate over time when preventive maintenance to unit j is performed only every τ_j time periods is shown in Figure 2.

The following constraints fully describe the value of the failure rate at any time t as a function of the preventive maintenance schedule.

$$\lambda_{jt} = \sum_{\theta=1}^{\tau_j} \gamma_{j\theta} Z_{jt\theta} \quad \forall j, t \tag{6}$$

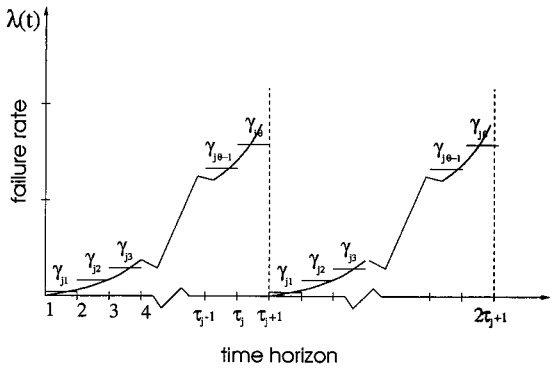


Figure 2. Equipment failure rate with preventive maintenance every τ_j periods.

$$Z_{jt\theta} \le X_{j,t-\theta} \quad \forall j, t, \theta = 1, \ldots, \tau_j \tag{7}$$

$$\sum_{\theta=1}^{\tau_j} Z_{jt\theta} = 1 \quad \forall j, t \tag{8}$$

It is also assumed that $X_{j0} = 1$, $\forall j$, and that $Z_{jt\theta}$, $\forall \theta > t$.

Constraint 7 allows for all of the $Z_{jt\theta}$ variables, for which a preventive maintenance action was last performed to unit j during period $t - \theta$, to take the value of 1. Because preventive maintenance is AGAN, only the most recent action must be considered when the value of the failure rate during period t is determined. This is enforced in two ways: (i) constraint 8 states that only one maintenance action determines the current value of the failure rate; and (ii) the optimization of the objective function, which involves the minimization of maintenance and lost production costs, favors a low failure rate value, and therefore, the most recent maintenance action determines the current value of the failure rate. Note that, according to constraint 6, it is not necessary to include the complete history of the failure rate but only its values in the last τ_j time periods. This is because at least one maintenance action is performed every τ_j time periods, thus restoring the equipment condition to AGAN status. The functionality of the failure rate constraints 6−8 is illustrated in Appendix A.

3.2. Derivation of Uptime Constraints. The expected equipment uptime is defined as the expected period of time during which an item is able to perform its intended function. Naturally, uptime depends on the expected number of failures and on the duration of maintenance, which are determined by the failure and maintenance characteristics of the equipment.

In the case of minimal repair and an AGAN preventive maintenance policy, the expected number of failures, $\bar{N}_{jT}$, of unit j operating for T time units is given[20] by

$$\bar{N}_{jT} = \int_0^T \lambda_j(s)\, ds \tag{9}$$

Three cases can be distinguished.[13]

Case 1. Equipment can fail during both minimal repair and preventive maintenance. In this case, the expected uptime for unit j during period t is given by

$$U_{jt} = H(1 - \Delta_j^c \lambda_{jt}) - \Delta_j^p X_{jt} \quad \forall j, t \tag{10}$$

The above equation simply states that the uptime, U_{jt},

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is equal to the interval duration, H, minus the repair time, $H\Delta_j^c\lambda_{jt}$, minus the time dedicated to preventive maintenance, $\Delta_j^p X_{jt}$.

Case 2. Equipment can fail during minimal repair but *not* during preventive maintenance. In this case, the expected uptime for unit j during period t is given by

$$U_{jt} = (H - \Delta_j^p X_{jt})(1 - \Delta_j^c \lambda_{jt}) \quad \forall j, t \tag{11}$$

Note that the uptime for case 2 is larger than that for case 1 by the amount $\Delta_j^p X_{jt}\Delta_j^c\lambda_{jt}$ as there are no failures during preventive maintenance.

Case 3. No failures are possible during either minimal repair or preventive maintenance. In this case, the expected uptime for unit j during period t is given by

$$U_{jt} = (H - \Delta_j^p X_{jt})/(1 + \Delta_j^c \lambda_{jt}) \quad \forall j, t \tag{12}$$

It is clear that the uptime for case 3 is larger than those for the previous two cases as there is no failure during any maintenance activity.

These expressions relate the equipment uptime to the preventive maintenance policy (X_{jt}) and the equipment failure rate (λ_{jt}).

The presence of the equipment uptime in the resource ulitization constraints provides a link between the STN-based process planning model and the maintenance model proposed in section 3.1. Note, however, that expressions 11 and 12 are nonlinear. Therefore, before the maintenance and process models are combined into a simultaneous production planning and maintenance optimization formulation, appropriate linearizations are performed, as described in Appendix B, resulting in the following expressions

$$\overline{XZ}_{jt\theta} \le Z_{jt\theta} \quad \forall j, t, \theta = 1, ..., \tau_j \tag{13}$$

$$X_{jt} = \sum_{\theta=1}^{\tau_j} \overline{XZ}_{jt\theta} \quad \forall j, t \tag{14}$$

Case 2

$$U_{jt} = H - H\Delta_j^c \sum_{\theta=1}^{\tau_j} \gamma_{j\theta} Z_{jt\theta} - \Delta_j^p X_{jt} + \Delta_j^p \Delta_j^c \sum_{\theta=1}^{\tau_j} \gamma_{j\theta} \overline{XZ}_{jt\theta} \quad \forall j, t \tag{15}$$

Case 3

$$U_{jt} = \sum_{\theta=1}^{\tau_j} \frac{HZ_{jt\theta} - \Delta_j^p \overline{XZ}_{jt\theta}}{1 + \Delta_j^c \gamma_{j\theta}} \quad \forall j, t \tag{16}$$

4. A System Effectiveness Optimization Framework

In this section, an appropriate system effectiveness measure for multipurpose processes is first proposed, incorporating all operating and maintenance costs. Then, the maintenance model constraints and the aggregate planning model constraints described in the previous sections are combined to provide an optimization framework for the simultaneous identification of maintenance and production plans.

The additional notation used is listed below:

Parameters

η_{st} = unit price of state s during period t

C_{ut} = unit cost of utility u during period t

C_{jt}^p = preventive maintenance cost of unit j during period t

C_{jt}^c = corrective maintenance cost of unit j during period t

The following expression for the expected profit is used as our system effectiveness measure:

$$\Phi = \sum_{st} \eta_{st} D_{st} - \sum_{ut} C_{ut} \sum_i \sum_{j \in K_i} \sum_{\omega=0}^{p_i - 1} (\beta_{uij\omega} N_{ijt} + \delta_{uij\omega} B_{ijt}) - \sum_{jt} C_{jt}^p X_{jt} - \sum_{jt} C_{jt}^c (H - U_{jt} - \Delta_j^p X_{jt})/\Delta_j^c \tag{17}$$

In eq 17, the first term represents the profit generated by the delivered products, while the second term denotes the cost of utilities. The third term corresponds to the cost of preventive maintenance activities, expressed as a linear function of X_{jt} variables. Finally, the fourth term is related to the corrective maintenance cost, with the term $(H - U_{jt} - \Delta_j^p X_{jt})$ denoting the total time dedicated to repairs. (Thus, the number of corrective maintenance activities is simply given by dividing this term by the duration of each repair, Δ_j^c.)

Note that the use of Φ as the overall system effectiveness criterion is fully justified, as it provides the optimal balance between process revenues, operating costs, and maintenance costs (see also refs 10 and 11)

By combining all of the appropriate terms and expressions, derived in previous sections, the problem of identifying the optimal production and maintenance plan, having taken into consideration their interactions, corresponds to the following optimization problem (P1).

Problem P1

$$\max \Phi = \sum_{st} \eta_{st} D_{st} - \sum_{ut} C_{ut} \sum_i \sum_{j \in K_i} \sum_{\omega=0}^{p_i - 1} (\beta_{uij\omega} N_{ijt} + \delta_{uij\omega} B_{ijt}) - \sum_{jt} C_{jt}^p X_{jt} - \sum_{jt} C_{jt}^c (H - U_{jt} - \Delta_j^p X_{jt})/\Delta_j^c$$

subject to

Resource Utilization Constraints

$$\sum_{i \in I_j} p_i N_{ijt} \le U_{jt} \quad \forall j, t$$

Capacity Constraints

$$\phi_{ij}^{\min} V_j N_{ijt} \le B_{ijt} \le \phi_{ij}^{\max} V_j N_{ijt} \quad \forall i, j \in K_i, t$$

Material Balances

$$S_{st} = S_{s,t-1} + \sum_{i \in T_s} \sum_{j \in K_i} \bar{\rho}_{is} B_{ijt} - \sum_{i \in T_s} \sum_{j \in K_i} \rho_{is} B_{ijt} - D_{st} \quad \forall s, t$$

Demand Constraints

$$D_{st}^{\min} \le D_{st} \le D_{st}^{\max} \quad \forall s, t$$

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Utility Constraints

$$\sum_i \sum_{j \in K_i} \sum_{\omega=0}^{p_i-1} \beta_{uij\omega} N_{ijt} + \delta_{uij\omega} B_{ijt} \le A_{ut}^{\max} H \quad \forall u, t$$

Uptime Definition Constraints

$$\text{Case 1} \qquad U_{jt} = H(1 - \Delta_j^{\mathrm{c}} \lambda_{jt}) - \Delta_j^{\mathrm{p}} X_{jt} \quad \forall j, t$$

$$\text{Case 2} \qquad U_{jt} = (H - \Delta_j^{\mathrm{p}} X_{jt})(1 - \Delta_j^{\mathrm{c}} \sum_{\theta=1}^{\tau_j} \gamma_{j\theta} Z_{jt\theta}) \quad \forall j, t$$

$$\text{Case 3} \qquad U_{jt} = \sum_{\theta=1}^{\tau_j} \frac{H Z_{jt\theta} - \Delta_j^{\mathrm{p}} \overline{XZ}_{jt\theta}}{1 + \Delta_j^{\mathrm{c}} \gamma_{j\theta}} \quad \forall j, t$$

Linearization Constraints (for cases 2 and 3)

$$\overline{XZ}_{jt\theta} \le Z_{jt\theta} \quad \forall j, t, \theta = 1, \ldots, \tau_j$$

$$X_{jt} = \sum_{\theta=1}^{\tau_j} \overline{XZ}_{jt\theta} \quad \forall j, t$$

Failure Rate Constraints

$$\lambda_{jt} = \sum_{\theta=1}^{\tau_j} \gamma_{j\theta} Z_{jt\theta} \quad \forall j, t$$

$$Z_{jt\theta} \le X_{j,t-\theta} \quad \forall j, t, \theta = 1, \ldots, \tau_j$$

$$\sum_{\theta=1}^{\tau_j} Z_{jt\theta} = 1 \quad \forall j, t$$

Note that, in case 1, the uptime definition constraints of eq 10 are used. To describe cases 2 and 3, the uptime definition constraints of eqs 15 and 16, respectively, are used, in conjunction with constraints 13 and 14. Also note that, in all three cases, problem P1 corresponds to a MILP model that can be solved using standard branch-and-bound techniques.

5. Simultaneous Design, Production, and Maintenance Planning

In this section, problem P1 is extended to form an optimization formulation for simultaneous design, production, and maintenance planning for multipurpose process plants. A summary of the proposed model was also briefly described in Pistikopoulos et al.[21]

The additional notation used in this section is listed below

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k = unit size

Set

Ψ_j = set of unit sizes available for unit j

Parameters

$\hat{V}_{jk}$ = size k for unit j

$N_{ijt}^{\max}$ = maximum number of batches when task i is performed on unit j during period t

K_j^0 = fixed cost for unit j

K_j^1 = variable size factor for unit j

together with the following variables:

Binary Variables

$E_j = 1$ if unit j is chosen; 0 otherwise

$\hat{E}_{jk} = 1$ if size k is chosen for unit j; 0 otherwise

Continuous Variable

V_j = size of unit j Next in this section, we describe how model P1 is extended to include design aspects as well.

5.1. Design Constraints. System structure—and the corresponding production recipes—is selected from a superstructure of units. The binary variable E_j is defined to denote whether unit j is selected to be part of the plant. Furthermore, the binary variable $\hat{E}_{jk}$ is defined to denote the size of unit j. In particular, unit j is assumed to be available at a finite number of known sizes. If a certain size k is chosen for unit j, then $\hat{E}_{jk} = 1$.

The following constraints state that, if unit j is selected to participate in the process, i.e., $E_j = 1$, then only one of the k possible sizes can be selected to represent the volume of that unit.

$$E_j = \sum_{k \in \Psi_j} \hat{E}_{jk}, \quad \forall j \tag{18}$$

$$V_j = \sum_{k \in \Psi_j} \hat{V}_{jk} \hat{E}_{jk}, \quad \forall j \tag{19}$$

5.2. Production Planning Constraints. The constraints in eq 3 are still used to describe the material balances. The capacity constraints (eq 2) combined with the constraints in eq 19 are now written as

$$\phi_{ij}^{\min} \sum_{k \in \Psi_j} \hat{V}_{jk} \hat{E}_{jk} N_{ijt} \le B_{ijt} \le \phi_{ij}^{\max} \sum_{k \in \Psi_j} \hat{V}_{jk} \hat{E}_{jk} N_{ijt} \quad \forall i, j \in K_i, t \tag{20}$$

Nonlinearities in the form of $\hat{E}_{jk} N_{ijt}$ can be linearized[22] by introducing a continuous positive variable, $\overline{EN}_{ijkt}$

$$\overline{EN}_{ijkt} \equiv \hat{E}_{jk} N_{ijt}, \quad \forall i, j \in K_i, k \in \Psi_j, t \tag{21}$$

together with the following constraints

$$\overline{EN}_{ijkt} \le N_{ij}^{\max} \hat{E}_{jk}, \quad \forall i, j \in K_i, k \in \Psi_j, t \tag{22}$$

$$N_{ijt} = \sum_k \overline{EN}_{ijkt} \quad \forall i, j \in K_i, t \tag{23}$$

where $N_{ij}^{\max}$ denotes the maximum number of batches of task i that can be produced in unit j.

Substituting eq 21 into constraint 20, the capacity constraints (eq 2) are now given by

$$\phi_{ij}^{\min} \sum_k \hat{V}_{jk} \overline{EN}_{ijkt} \le B_{ijt} \le \phi_{ij}^{\max} \sum_k \hat{V}_{jk} \overline{EN}_{ijkt} \quad \forall i, j \in K_i, t \tag{24}$$

The maximum number of batches is given by

$$N_{ij}^{\max} = \frac{U_j^{\max}}{p_i}, \quad \forall i, j \in K_i \tag{25}$$

Using the uptime expressions in eqs 10−12, the maximum expected uptime during each time period for cases 1 and 2 can be derived as

$$U_j^{\max} = H(1 - \Delta_j^{\mathrm{c}} \lambda_{j1}), \quad \forall j \tag{26}$$

whereas for case 3, it is given by

$$U_j^{\max} = \frac{H}{(1 + \Delta_j^{\mathrm{c}} \lambda_{j1})}, \quad \forall j \tag{27}$$

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5.3. Maintenance Modeling Constraints. To ensure that preventive maintenance activities for a unit j take place only if that unit exists, the following constraint is introduced

$$X_{jt} \le E_j, \quad \forall j, t \tag{28}$$

and constraint 8 is modified to

$$\sum_{\theta=1}^{\tau_j} Z_{jt\theta} = E_j \quad \forall j, t \tag{29}$$

The above constraints suggest that, if a unit j is not selected (i.e., $E_j = 0$), then no maintenance actions can be performed ($X_{jt} = 0$, $\forall t$, and $Z_{jt\theta} = 0$, $\forall t, \theta$). As a result, if a unit j is not selected to be part of the system, then, from constraint 6, its failure rate should always be zero ($\lambda_{jt} = 0$, $\forall t$) as all $Z_{jt\theta}$ variables are zero.

Furthermore, uptime constraints are modified as follows

Case 1

$$U_{jt} = H(E_j - \Delta_j^c \lambda_{jt}) - \Delta_j^p X_{jt} \quad \forall j, t \tag{30}$$

Case 2

$$U_{jt} = E_j H - H\Delta_j^c \sum_{\theta=1}^{\tau_j} \gamma_{j\theta} Z_{jt\theta} - \Delta_j^p X_{jt} + \Delta_j^p \Delta_j^c \sum_{\theta=1}^{\tau_j} \gamma_{j\theta} \overline{XZ}_{jt\theta} \quad \forall j, t \tag{31}$$

Case 3

No modification is needed

If a unit j is not selected, then these modifications, in conjunction with the fact that the linearization constraints in eqs 13 and 14 enforce that $\overline{XZ}_{jt\theta} = 0$, $\forall j, t, \theta$, suggest that its expected uptime is always zero ($U_{jt} = 0$, $\forall t$).

5.4. System Effectiveness Optimization. Each unit j is characterized by a fixed design cost, K_j^0, and cost per unit volume, K_j^1. Therefore, depending on whether unit j is part of the system or not, the total design cost is given by

$$\sum_j (K_j^0 E_j + K_j^1 \sum_{k \in \Psi_j} \hat{V}_{jk} E_{jk})$$

and the expected profit of the system now becomes

$$\Phi = \sum_{st} \eta_{st} D_{st} - \sum_{ut} C_{ut} \sum_i \sum_{j \in K_i} \sum_{\omega=0}^{p_i - 1} (\beta_{uij\omega} N_{ijt} + \delta_{uij\omega} B_{ijt}) - \sum_{jt} C_{jt}^p X_{jt} - \sum_{jt} C_{jt}^c (HE_j - U_{jt} - \Delta_j^p X_{jt})/\Delta_j^c - \sum_j (K_j^0 E_j + K_j^1 \sum_{k \in \Psi_j} \hat{V}_{jk} E_{jk}) \tag{32}$$

Note that the fourth term of the expected profit has been slighlty modified, with respect to eq 17, to ensure zero corrective maintenance costs for nonexisting units.

The overall problem is finally formulated as the MILP model P2, involving eq 32 as the objective function subject to the constraints in eqs 1, 3−8, 13, and 14 for cases 2 and 3, as well as those in eqs 18, 19, 22−24, 28, and 29, and either 30, 31, or 16.

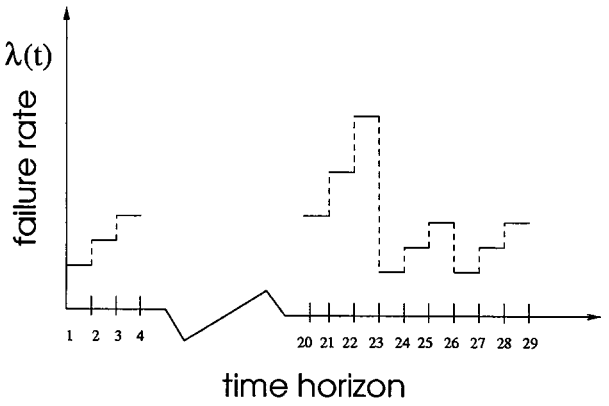


Figure 3. Failure rate profile.

Note that the results of the above optimal planning policy (obtained from the solution of problem P2) should be validated at the scheduling level (see, for example, refs 12 and 15). In that sense, the output of the planning model, such as the number and timings of the preventive maintenance tasks, can be incorporated into a detailed process/maintenance scheduling framework.

6. Numerical Examples

Two example problems are presented in this section. The first, which corresponds to a three-unit example, is used to demonstrate the key characteristics of our approach. The second is a large-scale example solved to depict the distinct computational advantages of the proposed methodology. All problems are modeled and solved within the GAMS modeling environment[23] using the CPLEX MILP optimizer.[24]

6.1. Example 1. Consider the process[13] described by the STN shown in Figure 4. The capacities of the units and storage facilities, as well as the types of tasks performed by each unit, are given in Table 1.

We consider an operating horizon of 2 years, 24 1-month periods. The demand for products B and C is between 5000 and 20 000 units for each month-long period with unit price per period of $\eta_{st} = 0.5$ for both B and C.

The failure rate and maintenance characteristics for all three units are as follows: $\tau_j = 9$, $\gamma_{j1} = 0.002\ \text{h}^{-1}$, $\gamma_{j\theta} = \gamma_{j,\theta-1} + 0.001\ \text{h}^{-1}$, $2 \le \theta \le \tau_j$, $\Delta_j^c = 24$ h, $\Delta_j^p = 6$ h, $C_j^c = 50$, $C_j^p = 1000$.

For case 1 (as described in section 3.2), the optimal preventive maintenance policy, by solving problem P1, is as follows: (i) Unit 1 is preventively maintained at $t = 8$ and 16. (ii) Unit 2 is preventively maintained at $t = 3$, 6, 9, 12, 15, 18, and 21. (iii) Unit 3 is preventively maintained at $t = 4$, 8, 12, 16, and 20.

The corresponding failure rate profiles for the three units are shown in Figure 5. Note that, because unit 1 has a large capacity and can produce the required volumes of A during each period, a high expected uptime for unit 1 is not essential, and fewer preventive maintenance actions are scheduled. On the other hand, for units 2 and 3, which have smaller capacities, very high uptimes are required. Consequently, a large number of maintenance actions are required to produce the required volumes of B and C. Unit 2, in particular, is more important to the timely production of B and C as it has larger capacity than unit 3. Therefore, unit 2 is maintained more often than the other two units.

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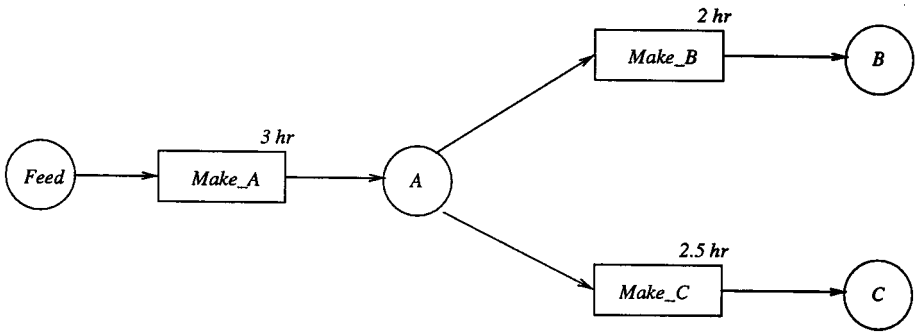


Figure 4. STN for example 1.

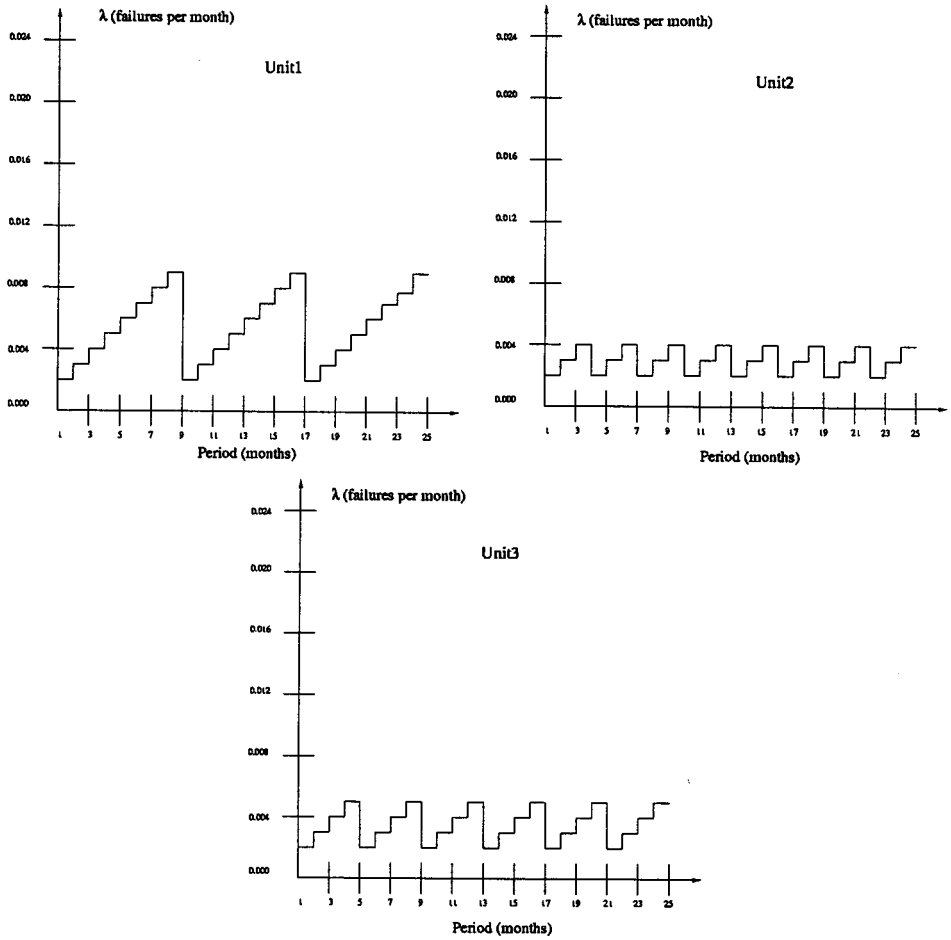


Figure 5. Optimal failure rate profiles: Example 1, case 1.

Table 1. Details of Resources: Example 1

unit	capacity	suitability
unit1	200	Make_A
unit2	50	Make_B, Make_C
unit3	40	Make_B, Make_C
Ftank	∞	feed
Btank	∞	B
Ctank	∞	C

The optimal maintenance policies in cases 2 and 3, as depicted by the solution of problem P1, exhibit trends similar to those in case 1, as shown in Figures 6 and 7. Again, unit 2 is maintained more often than unit 3, and unit 3 is maintained more often than unit 1. The slight differences in the maintenance schedules in the three cases are mainly due to different equipment failure behaviors during maintenance in each case.

The computational model statistics for the three MILP problems and the production and maintenance characteristics of the optimal policies in all cases are depicted in Tables 2 and 3, respectively. Note that, in all cases, the solution of the MILP problem was provided in one iteration by solving only the fully relaxed linear programming problem.

The solution obtained was then compared to two sequential strategies, in which production planning and maintenance planning are considered in sequence rather than in a simultaneous way through problem P1. In the first sequential strategy, a frequent maintenance policy for each unit is considered, whereas in the second scenario, a less frequent (infrequent) maintenance policy is adopted. Table 4 depicts the results compared to the solution of problem P1, case 1. Note that distinct economic benefits can be achieved if a simultaneuous strategy is adopted. It is also interesting to note that it is far from obvious what maintenance strategy to follow via a sequential approach, as the problem of identifying

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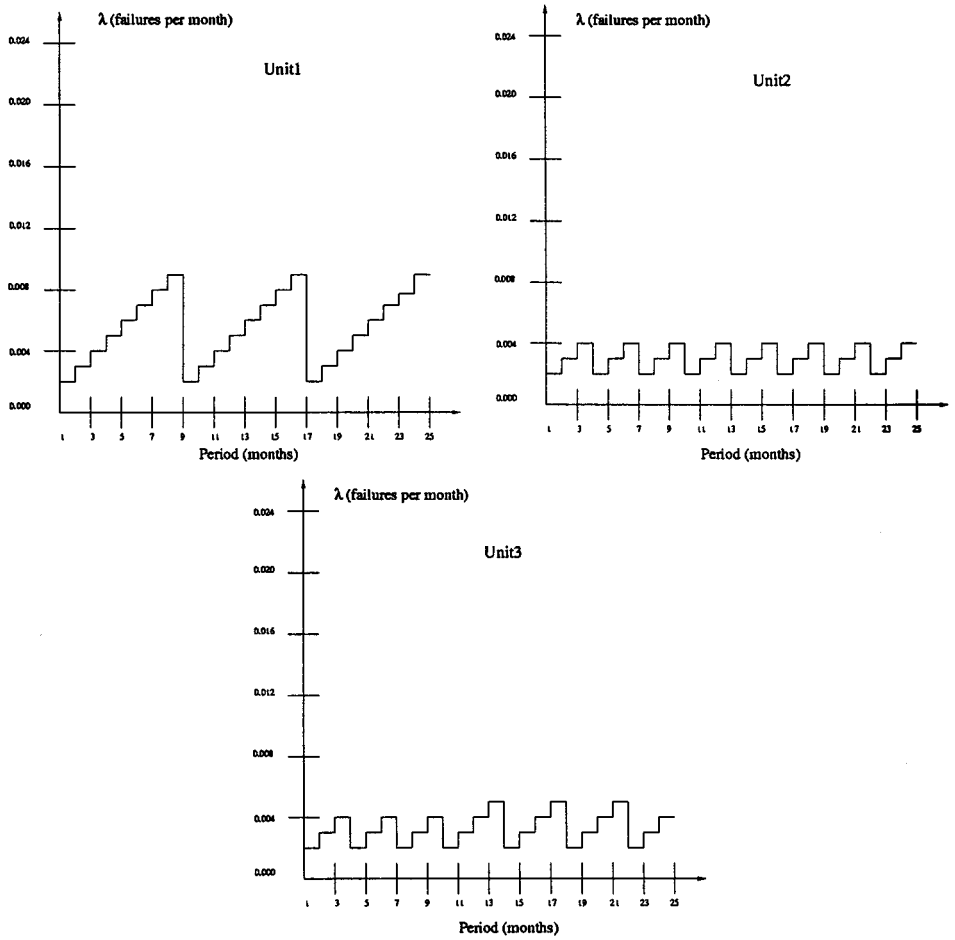


Figure 6. Optimal failure rate profiles: Example 1, case 2.

the right balance between an "optimal" maintenance plan and an "optimal" production plan becomes iterative and thus computationally expensive.

Finally, the case of simultaneous design, production, and maintenance planning was also considered. Equipment sizes were not fixed in this case, but they could take one value of the available discrete sizes depicted in Table 5. Additional design, failure rate, and maintenance data are given in Table 6.

The optimal equipment sizes, obtained from the solution of problem P2, are depicted in Table 7, whereas the resulting optimal maintenance policy is depicted in Figure 8. Finally, Table 8 presents the corresponding design and maintenance costs, as well the profitability characteristics, of these policies. The computational model statistics for the solution of problem P2 are shown in Table 9.

Note that a large equipment size is selected for unit 1. This is explained by the increased importance of unit 1 due to the performance of task Make_A (the production of intermediate product *A*), as well as the low design cost factors. Furthermore, unit 1 is maintained more often than the other two units despite the fact that it is not characterized by worse reliability characteristics than units 2 and 3. Again, the reason for this is the criticality of unit 1 in the production process because of the production of intermediate product *A*.

6.2. Example 2. Consider the process[25] described by the STN in Figure 9. The capacities of the units and storage facilities, the types of tasks performed by each unit, and the states stored in each storage unit are given in Tables 10 and 11.

The failure rate of the processing units is, as in the previous example, determined by the recursive relationship $\gamma_{j\theta} = \gamma_{j,\theta-1} + \alpha_j$, $\forall j$,. Furthermore, it is assumed that *at least* one maintenance action must be performed to each unit every 9 time periods, i.e., $\tau_j = 9$, $\forall j$. The necessary maintenance data are given in Table 12.

The operating horizon considered is 2 years, in 24 1-month periods. The demand for all products is between 250 and 800 units for each 1-month period, with the unit price per period of $\eta_{st} = 50$. Furthermore, it is assumed that the minimum and maximum capacity utilization factors are $\phi_{ij}^{\min} = 0.25$ and $\phi_{ij}^{\max} = 0.95$, respectively.

The optimal maintenance policy for the three cases considered is obtained by the solution of problem P1, the computational statistics of which are depicted in Table 13. The corresponding production and maintenance characteristics resulting from the optimal production and maintenance policies are depicted in Table 14.

The optimal preventive maintenance schedule in case 1 is shown in Figure 10. Note that all units are maintained twice, except for unit 2, which requires three maintenance actions. This is explained by (i) the small capacity of unit 2, which implies that a high availability is essential to perform the required number of processing tasks to meet the demand constraints; (ii) the low preventive maintenance cost of unit 2; and most importantly, (iii) the fact that failure to perform task T32, which can be performed by unit 2 and is involved in the production of all end products, can result in large production losses.

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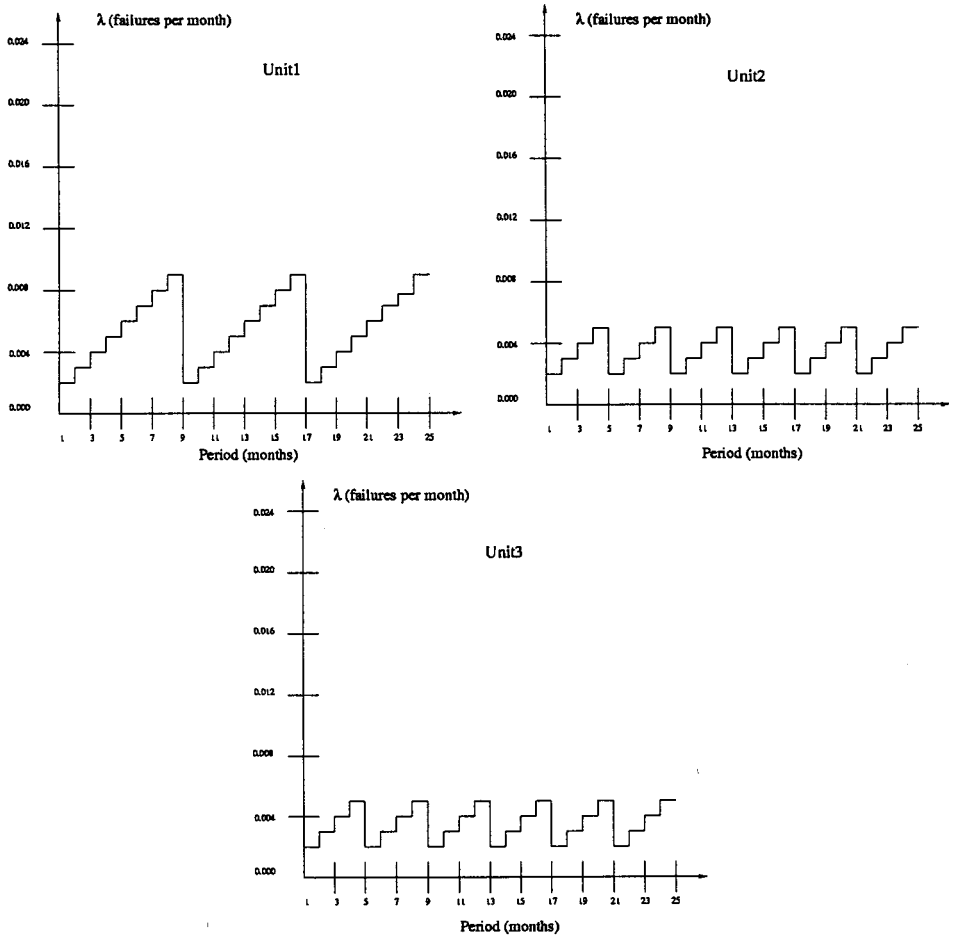


Figure 7. Optimal failure rate profiles: Example 1, case 3.

Table 2. Model P1 Statistics: Example 1

uptime model	number of binary variables	number of continuous variables	number of constraints	relative gap	CPU time[a]
case 1	612	409	901	0	2.3
case 2	612	949	1513	0	4.9
case 3	612	949	1513	0	5.8

[a] In units of seconds on a Sun SPARC 10 workstation.

Table 3. Deliveries and Maintenance Costs: Example 1

model	value of deliveries	total corrective maintenance cost	total preventive maintenance cost	objective function
case 1	334 326	10 368	14 000	309 958
case 2	335 179	10 130	15 000	310 049
case 3	334 631	9643	12 000	312 988

Table 4. Production and Maintenance Planning: Example 1

scenario	value of deliveries	total corrective maintenance cost	total preventive maintenance cost	objective function
frequent	339 080	6480	36 000	296 600
infrequent	317 660	14 256	9000	294 400
simultaneous (model P1)	334 326	10 368	14 000	309 958

Similar trends are observed in the optimal maintenance schedules of cases 2 and 3, which are identical to the maintenance schedule in case 1 (also depicted in Figure 10).

It is also interesting to note that, as shown in Table 13, the solution of the MILP problems (model P1), for all three cases, was obtained in a single iteration as the solution of the relaxed LP, similarly to example 1. Although nonconclusive at this stage, the zero integrality gap of the relaxed LP can be attributed to the unimodularity of the matrix of problem P1, an issue that is currently under theoretical investigation.

Table 5. Design Alternatives: Example 1

unit type	available unit sizes			
unit1	150	175	200	250
unit2	50	80	150	200
unit3	60	100	125	200

Table 6. Design, Failure Rate, and Maintenance Data: Example 1

unit type	K_j^0	K_j^1	γ_{j1}	α_j	Δ^c (h)	C^c	Δ^p (h)	C^p
unit1	5000	100	0.002	0.001	24	50	6	1000
unit2	20 000	300	0.004	0.001	40	100	9	2000
unit3	20 000	350	0.002	0.001	30	75	7	2000

Table 7. Optimal Unit Sizes: Example 1

	case 1	case 2	case 3
unit1	250	250	250
unit2	80	200	60
unit3	125	50	200

7. Concluding Remarks

In this work, we have presented a novel system effectiveness optimization framework for analyzing and quantifying the interactions of process design, production planning, and maintenance planning in multipurpose process plants.

The key features of the proposed framework are (i) an analytical mathematical model for preventive maintenance of multipurpose equipment units, based on general piecewise-constant equipment failure rates; (ii) a multiperiod production planning model; (iii) explicit uptime constraints to quantify the availability of equip-

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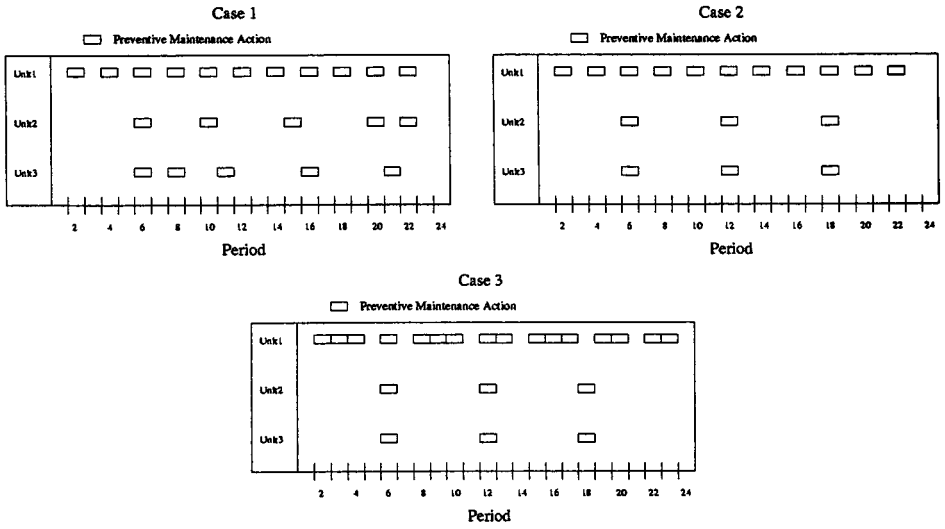


Figure 8. Optimal preventive maintenance schedule: Design optimization.

Table 8. Design, Deliveries, and Maintenance Costs: Example 1

uptime model	design cost	value of deliveries	total corrective maintenance cost	total preventive maintenance cost	objective function
case 1	137 750	674 050	17 208	24 000	495 092
case 2	151 000	674 253	19 114	20 000	484 139
case 3	155 000	678 229	15 662	25 000	482 567

Table 9. Model P2 Statistics: Example 1

uptime model	number of binary variables	number of constraint variables	number of constraints	relative gap	CPU time[a]
case 1	474	1011	1545	0.03	71.0
case 2	474	1398	2004	0.03	162.6
case 3	474	1398	2004	0.04	138.7

[a] In units of seconds on a Sun SPARC 10 workstation.

ment in the case of repair and preventive maintenance; and (iv) an expected profit objective function, which balances the process design/production costs with the maintenance costs, thereby corresponding to a system effectiveness criterion.

The resulting optimization problems correspond to MILP formulations, which require very modest computational effort for their solution, as demonstrated by the numerical examples presented. Other types of maintenance models, such as Markov-based problems, can also be included in such a system effectiveness framework, as discussed, for example, in ref 26.

Appendix A. Failure Rate Constraints—Illustrative Examples

The functionality of the failure rate constraints in eqs 6−8 is illustrated with three examples.

A.1. Example 1. Assume that we want to determine the value of the failure rate of unit j at time period $t = 25$, which was last maintained at time period $t = 22$ (see Figure 3). Also, assume that unit j has to be preventively maintained at least once every $\tau_j = 5$ time periods.

By applying constraint 6, we obtain

$$\lambda_{j25} = \gamma_{j1}Z_{j25,1} + \gamma_{j2}Z_{j25,2} + \gamma_{j3}Z_{j25,3} + \gamma_{j4}Z_{j25,4} + \gamma_{j5}Z_{j25,5}$$

By applying constraint 7, we obtain

$$Z_{j25,1} \le X_{j24}$$
$$Z_{j25,2} \le X_{j23}$$
$$Z_{j25,3} \le X_{j22}$$
$$Z_{j25,4} \le X_{j21}$$
$$Z_{j25,5} \le X_{j20}$$

Because only $X_{j22} = 1$, it is suggested that $Z_{j25,3}$ is the *only* $Z_{j25\theta}$ value that can be nonzero. Furthermore, by applying constraint 8

$$Z_{j25,1} + Z_{j25,2} + Z_{j25,3} + Z_{j25,4} + Z_{j25,5} = 1$$

it is suggested that $Z_{j25,3} = 1$. Therefore, the value of the failure rate at $t = 25$ is $\lambda_{j25} = \gamma_{j3}$, as determined by constraint 6.

A.2. Example 2. Assume that the value of the failure rate at $t = 27$ is required. Note that, during the previous $\tau_j = 5$ periods, unit j has been preventively maintained twice, for example, at $t = 22$ and at $t = 25$. The correct value of the failure rate at $t = 27$ should be determined only by the last maintenance action (i.e., $t = 25$). By applying constraints 6−8, we have

$$\lambda_{j27} = \gamma_{j1}Z_{j27,1} + \gamma_{j2}Z_{j27\,2} + \gamma_{j3}Z_{j27,3} + \gamma_{j4}Z_{j27,4} + \gamma_{j5}Z_{j27,5}$$

$$Z_{j27,1} \le X_{j26}$$
$$Z_{j27,2} \le X_{j25}$$
$$Z_{j27,3} \le X_{j24}$$
$$Z_{j27,4} \le X_{j23}$$
$$Z_{j27,5} \le X_{j22}$$
$$Z_{j27,1} + Z_{j27,2} + Z_{j27,3} + Z_{j27,4} + Z_{j27,5} = 1$$

Because both X_{j22} and X_{j25} equal one, constraints 7 and 8 state that only one of $Z_{j27,5}$ and $Z_{j27,2}$ can have the value of one. The required minimization of maintenance costs and maximization of expected uptime for j will enforce constraint 6 to assign the lowest possible value

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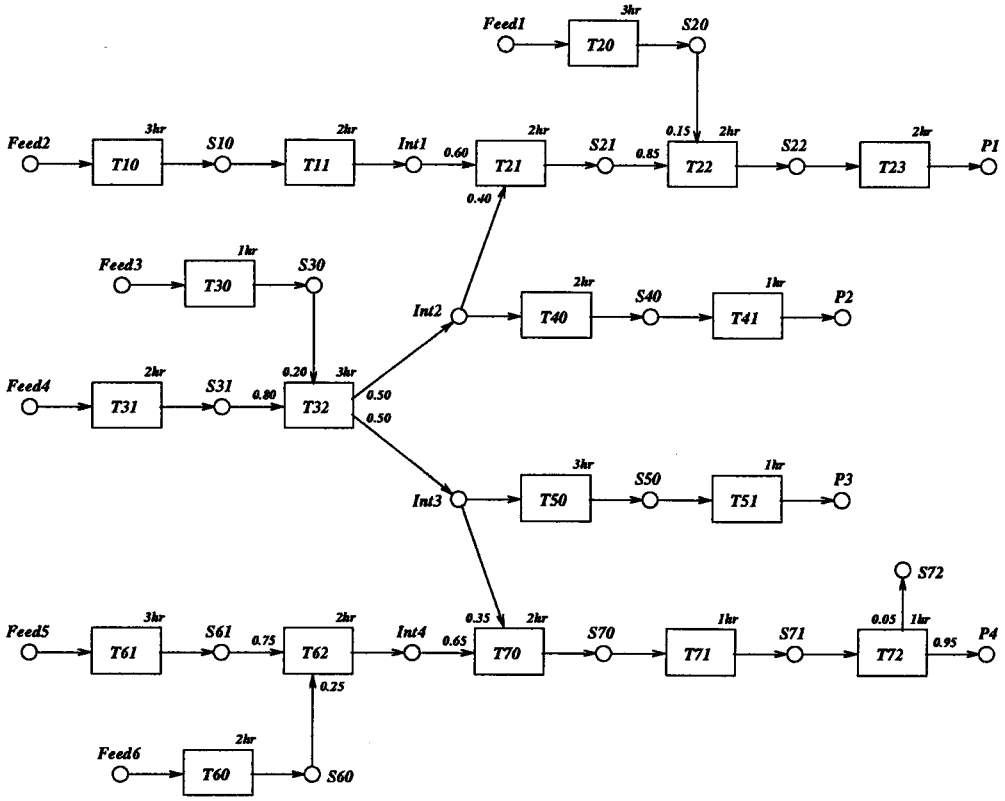


Figure 9. STN for example 2.

Table 10. Details of Processing Resources: Example 2

unit type	capacity (t)	suitable tasks
unit1	6.0	T10, T21
unit2	5.0	T32, T41, T51
unit3	7.0	T31, T72
unit4	7.0	T23, T60, T30
unit5	8.0	T40, T50, T20
unit6	6.0	T61, T70
unit7	7.0	T11, T22, T41
unit8	8.0	T62, T71, T51, T72

Table 11. Details of Storage Resources: Example 2

storage unit	capacity (t)	suitable states
warehouse	∞	Feed1−Feed6
tank1	30	int1
tank2	30	int2
tank3	30	int3
tank4	30	int4
warehouse	∞	P1−P4

Table 12. Failure Rate and Maintenance Data: Example 2

unit type	γ_{j1}	α_j	Δ^c (h)	C^c	Δ^p (h)	C^p
unit1, unit2	0.0007	0.0002	20	2000	15	10 000
unit3, unit4	0.0012	0.0002	25	2500	20	15 000
unit5, unit6	0.0015	0.0003	30	3000	25	20 000
unit7, unit8	0.0025	0.0003	35	3500	30	25 000

to the equipment failure rate. Therefore, $Z_{j27,2} = 1$ and $\lambda_{j27} = \gamma_{j2}$, which corresponds to the correct failure rate value.

Appendix B. Equipment Uptime Constraints—Linearization

The linearization constraints applied to cases 2 and 3 of section 3.2 are provided next.

B.1. Case 2. Substituting the failure rate expression in eq 6 into the uptime constraint 11, the expected equipment uptime is now given as a function of the 0−1 variables ($Z_{jt\theta}$, X_{jt})

$$U_{jt} = (H - \Delta_j^p X_{jt})(1 - \Delta_j^c \sum_{\theta=1}^{\tau_j} \gamma_{j\theta} Z_{jt\theta}) \quad \forall j, t$$

or

$$U_{jt} = H - H\Delta_j^c \sum_{\theta=1}^{\tau_j} \gamma_{j\theta} Z_{jt\theta} - \Delta_j^p X_{jt} + \Delta_j^p \Delta_j^c \sum_{\theta=1}^{\tau_j} \gamma_{j\theta} X_{jt} Z_{jt\theta} \quad \forall j, t \quad (33)$$

Note that the last term in the above expression comprises a summation of nonlinear products $X_{jt}Z_{jt\theta}$. To linearize terms of such a form, the new continuous variable $\overline{XZ}_{jt\theta}$ is introduced[22]

$$\overline{XZ}_{jt\theta} \equiv X_{jt} Z_{jt\theta} \quad \forall j, t, \theta = 1, ..., \tau_j$$

together with the constraints

Table 13. Model P1 Statistics: Example 2

up-time model	number of binary variables	number of constraint variables	number of constraints	relative gap	CPU time[a]	objective function
case 1	1632	2401	3745	0	14.8	1 333 713
case 2	1632	3841	5377	0	21.09	1 337 645
case 3	1632	3841	5377	0	18.16	1 417 226

[a] In units of seconds on a Sun SPARC 10 workstation.

Table 14. Deliveries and Maintenance Costs: Example 2

scenario	value of deliveries	total corrective maintenance cost	total preventive maintenance cost	objective function
case 1	2 571 521	947 808	290 000	1 333 713
case 2	2 571 642	943 997	290 000	1 337 645
case 3	2 572 468	865 242	290 000	1 417 226

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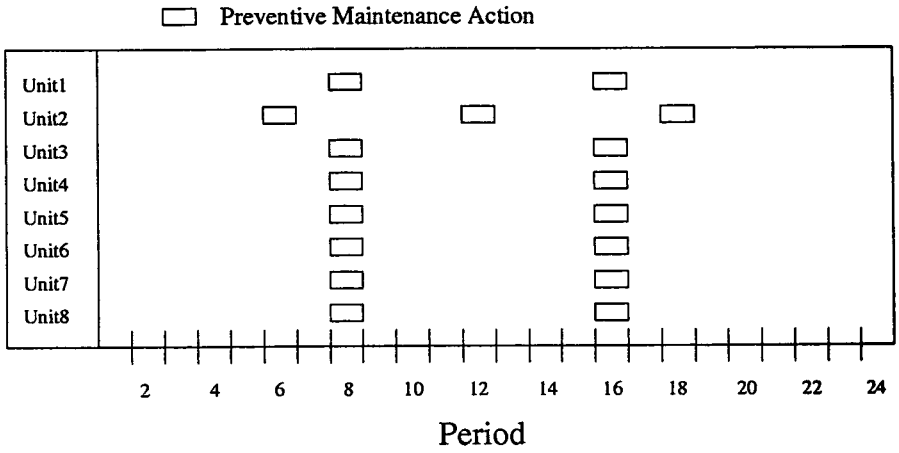


Figure 10. Optimal preventive maintenance schedule: Example 2, case 1.

$$\overline{XZ}_{jt\theta} \le Z_{jt\theta} \quad \forall j, t, \theta = 1, ..., \tau_j \tag{34}$$

$$X_{jt} = \sum_{\theta=1}^{\tau_j} \overline{XZ}_{jt\theta} \quad \forall j, t \tag{35}$$

By substituting $\overline{XZ}_{jt\theta}$, expression 33 can be rewritten as

$$U_{jt} = H - H\Delta_j^c \sum_{\theta=1}^{\tau_j} \gamma_{j\theta} Z_{jt\theta} - \Delta_j^p X_{jt} + \Delta_j^p \Delta_j^c \sum_{\theta=1}^{\tau_j} \gamma_{j\theta} \overline{XZ}_{jt\theta} \quad \forall j, t \tag{36}$$

The expected equipment uptime in the case where failure can occur during minimal repair but not during preventive maintenance is now fully described by the linear constraints in eqs 13−15.

B.2. Case 3. Substituting the failure rate expression 6 into the uptime constraint in eq 12, the expected equipment uptime now becomes

$$U_{jt} = \frac{H - \Delta_j^p X_{jt}}{1 + \Delta_j^c \sum_{\theta=1}^{\tau_j} \gamma_{j\theta} Z_{jt\theta}} = \frac{H - \Delta_j^p X_{jt}}{1 + \Delta_j^c \sum_{\theta=1}^{\tau_j} \gamma_{j\theta} Z_{jt\theta}} \quad \forall j, t \tag{37}$$

Taking into account constraint 8, which suggests that only one of the $Z_{jt\theta}$ variables in the above expression can take the value of 1, expression 37 is equivalent to

$$U_{jt} = \sum_{\theta=1}^{\tau_j} \frac{HZ_{jt\theta} - \Delta_j^p X_{jt} Z_{jt\theta}}{1 + \Delta_j^c \gamma_{j\theta}} \quad \forall j, t \tag{38}$$

The nonlinear terms $X_{jt}Z_{jt\theta}$ are linearized in the same manner as before. Then, expression 38 is rewritten as

$$U_{jt} = \sum_{\theta=1}^{\tau_j} \frac{HZ_{jt\theta} - \Delta_j^p \overline{XZ}_{jt\theta}}{1 + \Delta_j^c \gamma_{j\theta}} \quad \forall j, t \tag{39}$$

Therefore, the expected equipment uptime in the case where no failures are possible either during minimal repair or during preventive maintenance is now fully described by the linear constraints in eqs 13, 14, and 16.

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